Seraphina Sterling (617) 555-1212 | seraphina.sterling@email.com | linkedin.com/in/seraphinasterling | github.com/seraphinasterling

Professional Summary

Highly accomplished and results-oriented Lead Machine Learning Scientist with 8+ years of experience leading and contributing to the development and deployment of cutting-edge AI/ML solutions at Quantum Horizon Corp. Proven ability to translate complex business problems into innovative machine learning models, resulting in significant improvements in efficiency, accuracy, and profitability. Expertise in a wide range of machine learning techniques, including deep learning, natural language processing, and computer vision. Strong leadership and communication skills, with a demonstrated ability to mentor and guide junior team members. Passionate about leveraging AI to solve real-world problems and drive positive impact.

Education

Massachusetts Institute of Technology (MIT), Cambridge, MA * Doctor of Philosophy (Ph.D.), Electrical Engineering and Computer Science, May 2015 * Dissertation: "Novel Architectures for Deep Reinforcement Learning in Robotics" * Master of Science (M.S.), Electrical Engineering and Computer Science, May 2013 * Bachelor of Science (B.S.), Electrical Engineering and Computer Science, May 2011

Technical Skills

- **Programming Languages:** Python (Pandas, NumPy, SciPy), R, Java, C++
- **Machine Learning Libraries:** Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost
- **Cloud Computing:** Google Cloud Platform (GCP) (Compute Engine, Cloud Storage, BigQuery, Dataflow), AWS (S3, EC2), Azure
- **Databases:** SQL, NoSQL (MongoDB, Cassandra)
- **Big Data Technologies:** Hadoop, Spark
- **Data Visualization:** Matplotlib, Seaborn, Tableau
- **Version Control:** Git

Professional Experience

Lead Machine Learning Scientist, Quantum Horizon Corp., Boston, MA | June 2017 – Present

- Led a team of 5 machine learning engineers in the development and deployment of a real-time fraud detection system, resulting in a 25% reduction in fraudulent transactions.
- Designed and implemented a novel deep learning model for natural language processing, improving the accuracy of sentiment analysis by 15%.

- Developed and deployed a scalable machine learning pipeline using GCP, processing over 1 terabyte of data per day.
- Mentored and trained junior data scientists in best practices for machine learning model development and deployment.
- Successfully secured $2M in funding for new AI/ML initiatives.
- Presented research findings at several international conferences.

Machine Learning Engineer, NovaTech Solutions, Cambridge, MA | June 2015 – June 2017

- Developed and deployed machine learning models for various clients, including financial institutions and healthcare companies.
- Implemented machine learning algorithms for predictive maintenance, resulting in a 10% reduction in equipment downtime.
- Designed and implemented data pipelines for processing large datasets.
- Collaborated with cross-functional teams to integrate machine learning models into existing systems.

AI/ML Projects

- **Real-time Fraud Detection System (Quantum Horizon Corp.):** Developed a real-time fraud detection system using a combination of supervised and unsupervised learning techniques. The system achieved 98% accuracy in detecting fraudulent transactions. Utilized TensorFlow and GCP for model training and deployment. (Detailed technical report available upon request)

- **Sentiment Analysis Model (Quantum Horizon Corp.):** Developed a novel deep learning model for sentiment analysis, achieving state-of-the-art results on several benchmark datasets. The model utilized a combination of recurrent neural networks and convolutional neural networks. (GitHub repository available upon request)

- **Predictive Maintenance Model (NovaTech Solutions):** Developed a predictive maintenance model for industrial equipment using time series analysis and machine learning techniques. The model accurately predicted equipment failures, resulting in a significant reduction in downtime. (Confidential client data)

- **Image Recognition System (MIT Ph.D. Dissertation):** Developed a novel architecture for deep reinforcement learning in robotics, enabling robots to learn complex manipulation tasks through interaction with the environment. (Published in IEEE Robotics and Automation Letters)